GM	WORLDWIDE ENGINEERING STANDARDS	Test Procedure Noise and Vibration	GMW14876

Tire Force Transmissibility and Wheel Lateral Stiffness Evaluation Procedure

1 Scope

The Hand Held Force Input Frequency Response Function (FRF) Acquisition Master Test Procedure, GMW14181 and the Transducer Calibration Master Test Procedure, GMW14172, shall be referenced for more detailed information corresponding to the following procedure.

Note: Nothing in this standard supersedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. Define the process for obtaining the tire and wheel force transmissibility and wheel lateral stiffness. These measurements are used to compare tire force transmissibility and wheel lateral stiffness to the specifications.

1.2 Foreword. The tire and wheel force transmissibility and wheel lateral stiffness is a factor in structure borne road noise performance.

1.3 Applicability. This test procedure is appropriate to use for all passenger cars, vans and small bus tire and wheel systems.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

None

2.2 GM Standards/Specifications.

GMW14181	GMW14172
GMW8702	

2.3 Additional References.

3 Resources

3.1 Facilities.

A test room or an area which is free of extraneous vibration (per Hand Held Force Input Frequency Response Function (FRF) Acquisition Master Test Procedure, GMW14181).

3.2 Equipment.

3.2.1 Test Equipment.

Two measuring methods can be used to measure the tire and wheel transmissibility under free-free condition: Excitation with impact hammer or hand held shaker. The required equipment for both methods is different, the measurement results are comparable. Equivalent measurement equipment may be used in place of the suggested equipment.

3.2.1.1 Equipment (excitation with impact hammer)

- 4 channel FFT analyzer (Example: OROS25)
- Impact hammer with force transducer and nylon hammer tip (Example: PCB Small Impact Hammer)
- One triaxial accelerometer for vertical and lateral acceleration on the tire patch (Example: PCB 354 B33 ICP)
- Two single axis accelerometers for acceleration on each side of wheel center (Example: PCB 353 B65 ICP)
- Accelerometer Calibrator
- Calibration Mass

Note: Refer to GMW 14172 for more calibration details

3.2.1.2 Equipment (excitation with hand held shaker)

- 4 channel FFT analyzer (Example: OROS 25)
- Waveform generator (Example: WAVETEK Model 395)
- Hand held shaker with force transducer (Example Shaker: B&K Type 4819 and Force Transducer: B&K B200 185 524 60)
- Power Amplifier (Example: B&K 2706)
- Charge Amplifier (Example: B&K Typ 2635)
- One triaxial accelerometer for vertical and lateral acceleration on the tire patch (Example: PCB 354 B33 ICP)
- Two single axis accelerometers for acceleration on each side of wheel center (Example: PCB 353 B65 ICP)
- Accelerometer Calibrator
- Calibration Mass

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Note: Refer GMW 14172 for more calibration details

Note: Alternative equipment by other manufacturers are useable.

3.3 Test Vehicle/Test Piece.

Tire and wheel.

3.4 Test Time.

Calendar time: 0.5	days
Test hours: 2	hours
Coordination hours: 2	hours

3.5 Test Required Information.

3.6 Personnel/Skills. At least one technician or engineer skilled in noise and vibration data acquisition, analysis, equipment use, and test setup is required.

4 Procedure

4.1 Preparation.

4.1.1 The test tire and wheel will be hung up from elastic cords to allow the free-free condition in the direction of excitation. The rigid body mode of the tire and wheel shall be below 3 Hz. The test tire and wheel shall be secured appropriately to the elastic cords and the loading limits of the elastic cords shall not be exceeded. (See Appendix A for example of tire and wheel boundary condition)

4.1.2 The single axis accelerometers will be fixed on each side of the wheel center with hot glue in the direction of excitation. (see Appendix A for accelerometer placement)

4.1.3 The triaxial accelerometer will be fixed directly on the tire tread with hot glue. Only the vertical and lateral directions will be used. (see Appendix A for accelerometer placement).

4.1.4 For accelerometer and force transducer calibrations see GMW14172 Transducer Calibration Master Test Procedure.

4.2 Conditions.

4.2.1 Environmental Conditions.

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 Impact hammer excitation

4.3.1.1 Analyzer Acquisition Parameters

- CH1: Force signal Impact Hammer AC coupled auto range
- CH2: Acceleration signal tire tread (M3) ICP coupled auto range
- CH3: Acceleration signal wheel below center (M1) ICP coupled auto range
- CH4: Acceleration signal wheel above center (M2) ICP coupled auto range
- Trigger: 10% slope channel
- Average: 6 averages (min)
- Overlap none, rectangular window
- Max frequency: 2048 Hz
- Resolution: 1 Hz
- Enable overload detection

4.3.1.2 The excitation will be made with the impact hammer both above and below the wheel center in vertical direction.

Note: See GMW 14181 fore more details about excitation with impact hammer. Check whether the driving point accelerometer is fastened in vertical direction.

4.3.1.3 Recommended control signals on analyzer display while measurements:

a. FFT spectrum force input
b. Transfer FRFs
c. Driving point FRF
d. Coherence

4.3.1.4 Start the analyzer with PLAY.

4.3.1.5 Measurement will be stopped automatically after 6 averages

4.3.1.6 Store the measured data on hard disk.

a. Driving point FRFs (H11, H22)
b. Transfer FRFs (H31, H32, H21, H12)

4.3.1.7 Repeat the complete measurement with impact hammer excitation both above and below the wheel center in lateral direction and store measured data on hard disk.

Note: Check whether the driving point accelerometer is fastened in lateral direction.

4.3.2 Hand held shaker excitation

4.3.2.1 Select on waveform generator a sine signal (frequency 10-500 Hz). The frequency range should be continuously driven in UP/DOWN mode. The time duration for each run should be 15 seconds. Store the setup for the waveform generator attitudes.

4.3.2.2 Select at the power amplifier a force of +/- 1N.

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4.3.2.3 Select the right sensitivity of the force transducer at the charge amplifier (see force transducer documentation)

4.3.2.4 Analyzer Acquisition Parameters

- CH1: Force signal hand held shaker AC coupled (10 Hz high pass filter) auto range
- CH2: Acceleration signal tire tread (M3) ICP coupled auto range
- CH3: Acceleration signal wheel below center (M1) ICP coupled auto range
- CH4: Acceleration signal wheel above center (M2) ICP coupled auto range
- Average: 100 averages (linear)
- Overlap 50%, Hanning windows
- Max frequency: 500 Hz
- Resolution: 1 to 2 Hz
- Enable overload detection

4.3.2.5 The excitation will be made with the hand held shaker both above and below the wheel center in vertical direction.

Note: See GMW 14181 fore more details excitation with hand held shakers. Check whether the driving point accelerometer is fastened in vertical direction.

4.3.2.6 Recommended control signals on analyzer display while measurements:

a. All input time signals
b. Transfer FRFs
c. Driving point FRF
d. Coherence

4.3.2.7 Start the analyzer with PLAY.
4.3.2.8 Measurement will be stopped automatically after overall duration time.

4.3.2.9 Store the measured data on hard disk.

a. Driving point FRFs (H11, H22)
b. Transfer FRFs (H31, H32, H21, H12)

4.3.2.10 Repeat the complete measurement with hand held shaker excitation on wheel center in lateral direction and store measured data on hard disk. For the lateral measurement increase the analyzer max frequency to 2048 and increase the waveform generator to a maximum frequency of 1600 Hz.

Note: Check whether the driving point accelerometer is fastened in lateral direction.

5 Data

5.1 Calculations.

5.1.1 Drive Point and Transfer FRFs at wheel center.

Vertical FRFs

Drive Point FRF (at virtual wheel center)

$H_{00}=1/4(H_{11}+H_{21}-H_{12}-H_{22})$

Transfer FRF (from virtual wheel center to tire patch)

$H_{30}=1/2(H_{31}-H_{32})$

Lateral FRFs

Drive Point FRF (at virtual wheel center)

$H_{00}=1/4(H_{11}+H_{21}+H_{12}+H_{22})$

Transfer FRF (from virtual wheel center to tire patch)

$H_{30}=1/2(H_{31}+H_{32})$

5.1.2 Force Transmissibility

The force transmissibility FT in vertical and lateral will be calculated by the ratio:

$$FT = \frac{\text{Transfer FRF}}{\text{Driving point FRF}}$$

(Example see figure in Appendix B)

Data should be transferred into 1/3 octave spectrum. See Hand Held Force Input Frequency Response Function (FRF) Acquisition Master Test Procedure, GMW14181 Appendix C for calculation method.

(Example see figure in appendix B)

5.1.2 Lateral Wheel Stiffness

The lateral wheel stiffness can be calculated from the lateral drive point FRF H_{00}. This calculation is based on a three mass model in the case of the tire and wheel system and two mass model for the wheel only. The equations below yield results in units of N/m. Convert final result to units of kN/mm.

5.1.2.1 Lateral Wheel Stiffness for Tire and Wheel

$$k_{wheel} = (2 \cdot \pi \cdot f_2)^2 \left[M_T \left(\frac{f_4^{\,2}}{f_3^{\,2}} \right) - M_T \left(\frac{f_4^{\,2}}{f_3^{\,2}} \right) \left(\frac{f_2^{\,2}}{f_1^{\,2}} \right) \right]$$

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Where:
M_T = total system mass calculated using the average mass from 7-17 Hz (kg)
f_1 = lateral wheel mode (Hz)
f_2 = lateral wheel anti-resonance (Hz)
f_3 = wheel on tire mode (Hz)
f_4 = wheel on tire anti-resonance (Hz)

See Appendix C for example FRF.

5.1.2.2 Lateral Wheel Stiffness for Wheel Only

$$k_{wheel} = (2 \cdot \pi \cdot f_2)^2 \left[M_T - M_T \left(\frac{f_2^{\ 2}}{f_1^{\ 2}} \right) \right]$$

Where:
M_T = total system mass calculated using the average mass from 7-17 Hz (kg)
f_1 = lateral wheel mode (Hz)
f_2 = lateral wheel anti-resonance (Hz)

5.2 Interpretation of Results.

Verify whether SSTS requirements are achieved, when appropriate. Use the display format that facilitates this.

5.3 Test Documentation. The data are represented in a plotted form and must be present as data record readable with Excel.

All plots shall include the following information:

- Project and test name
- Test Engineer's name (or initials), phone and organization number.
- Date
- Type of measurement
- A brief description of the design level of the tires and wheels (mass of tire, mass of wheel ET)
- A more detailed description with data acquisition parameter shall be included in the test documentation.

Axes Labels shall be as follows:

x: Frequency: Hz. (range 50 to max usable frequency)

y: Amplitude with units (& dB reference)

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

7.2 Acronyms, Abbreviations, and Symbols.

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW14876

9 Release and Revisions

9.1 Release. This standard originated in September 2005. It was first approved by the Global N&V Structure-borne Noise Harmonization Team in November 2005. It was first published in November 2005.

9.2 Revisions.

Rev	Approval Date	Description (Organization)
A	Nov 2007	Revised procedure to eliminate center fixture
B	Nov 2008	Revise to include wheel stiffness calculation

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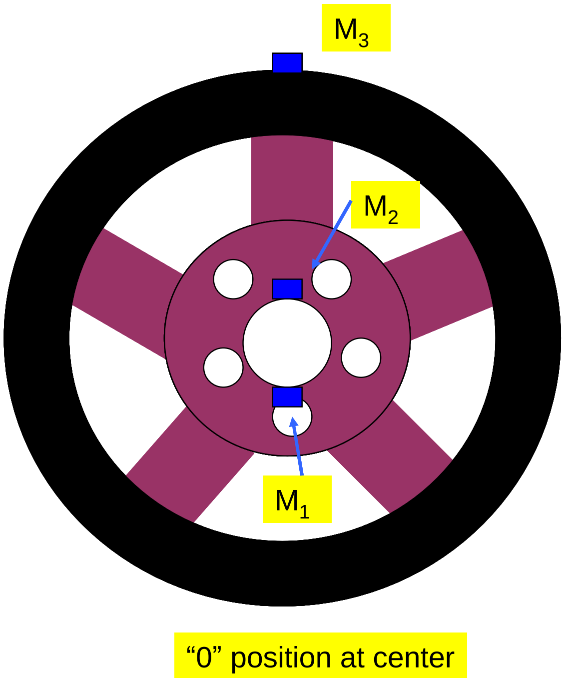
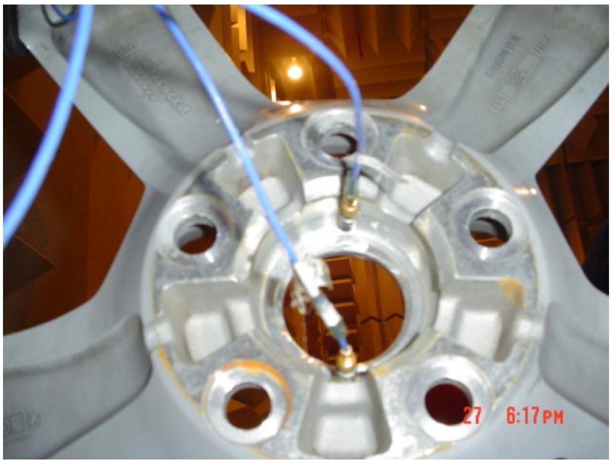
Appendix A

Tire and wheel setup

Free Free Mounting of tire and wheel on bungee cords



Placement of accelerometer on tire/wheel assembly



Placement of accelerometer at on each side of wheel center for lateral measurement. Orient accelerometers in vertical direction for vertical measurement

Triaxial Accelerometer on tire tread

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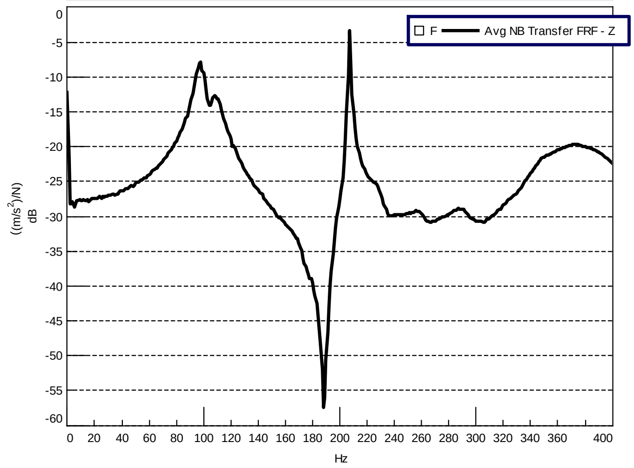
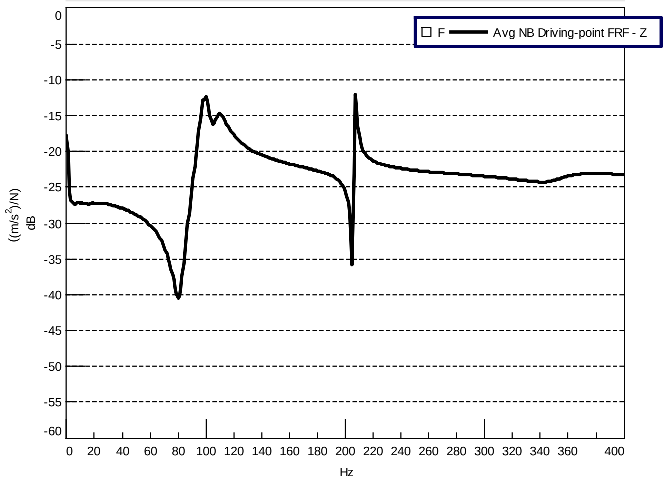
Appendix B

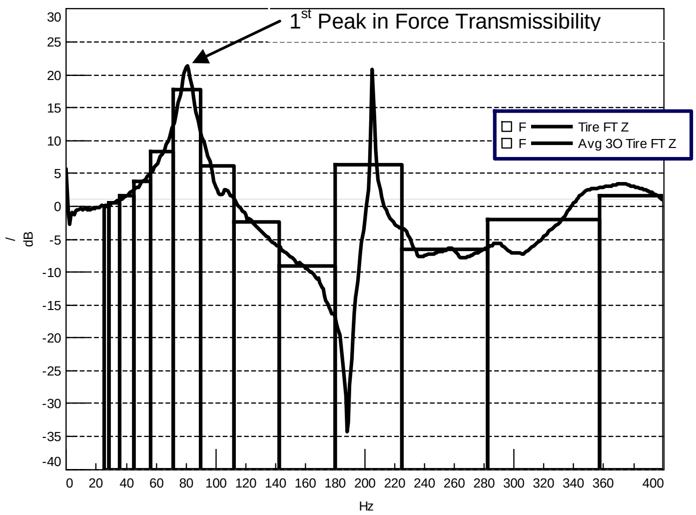
Example: Transfer FRF vertical direction



Example: Driving point FRF (wheel center) vertical direction



Example: Force Transmissibility calculation (Narrow band and 1/3 Octave)



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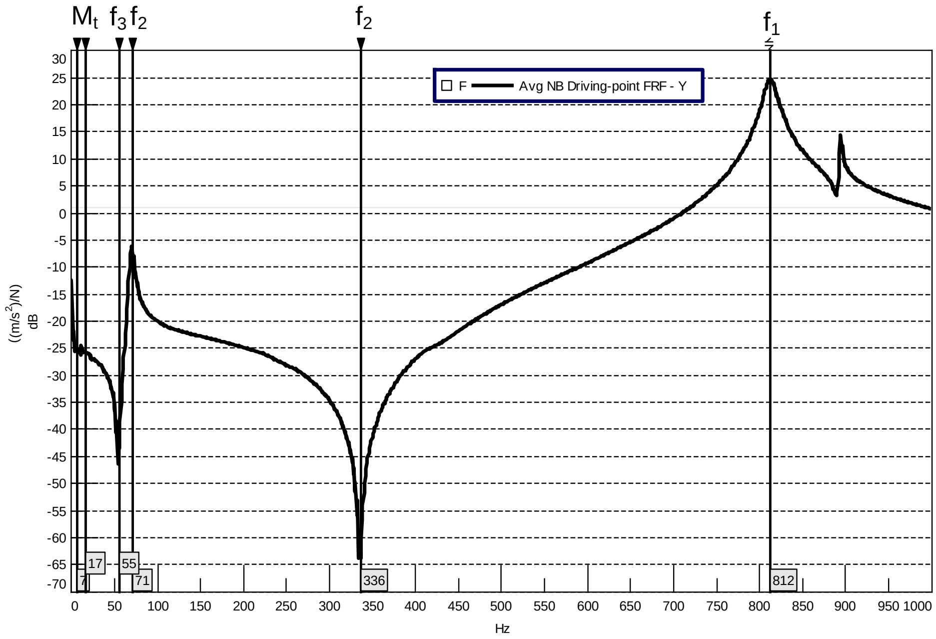
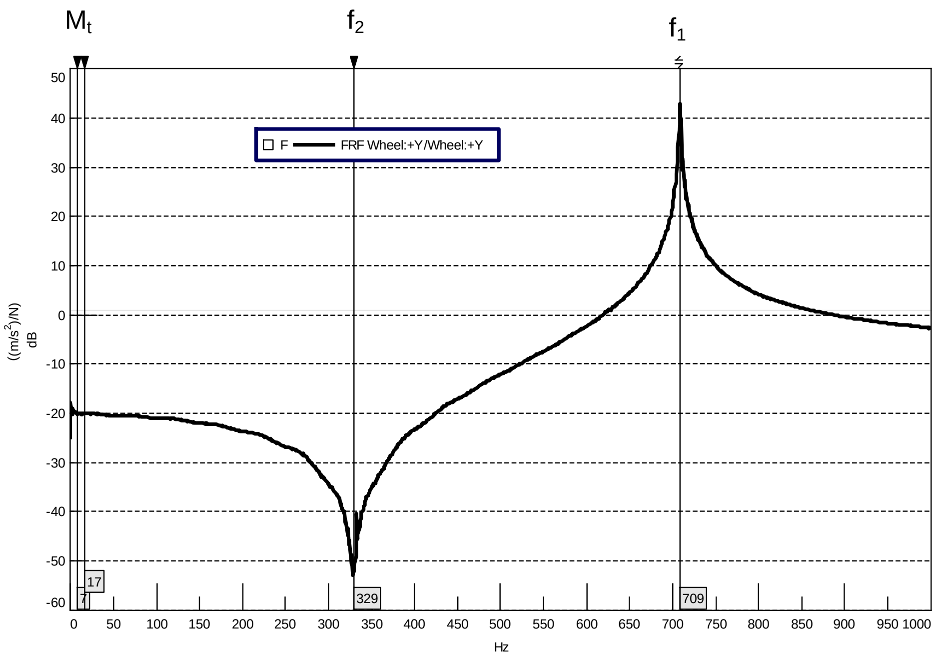
Appendix C

Example: Lateral Tire and Wheel Drive Point FRF



Example: Lateral Wheel Only Drive Point FRF



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